BRAC UNIVERSITY

Inspiring Excellence

Introduction to Transport Layer

Lecture 4 | CSE421 – Computer Networks

Department of Computer Science and Engineering
School of Data & Science

Our goals: Objectives

- understand principles behind transport layer services
- learn about two transport layer protocols:
  - UDP: User Datagram Protocol
  - TCP: Transmission Control Protocol

Transport vs. Network layer

- ***transport layer:*** logical communication between **processes**
- ***network layer:*** logical communication between **hosts**
- ***Segments:*** Transport Layer PDU

household analogy:

12 kids in Ann's house sending letters to 12 kids in Bill's house:

- processes = kids
- app messages = letters in envelopes
- hosts = houses
- transport protocol = Ann and Bill
- network-layer protocol = postal service

Functions of the Transport Layer

- Primary responsibilities:

1. Segmenting the data and managing each piece.
2. Reassembling the segments into streams of application data.
3. Identifying the different applications.
4. Multiplexing
5. Initiati[Reliability]ssion.
6. Performing flow control between end users.
7. Enabling error recovery.

Different Applications

Different Requirements

Web Applications













Online Games

- Some applications need their data to be complete with no errors or gaps.
- But can accept a slight delay to ensure this.

- Some applications can accept occasional errors or gaps in the data.
- But they cannot accept any delay.

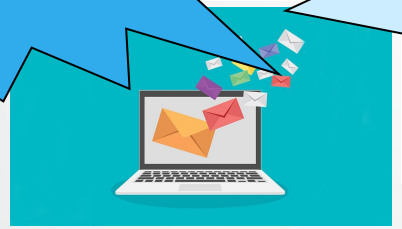
Solution : Two transport protocols?

Web Applications

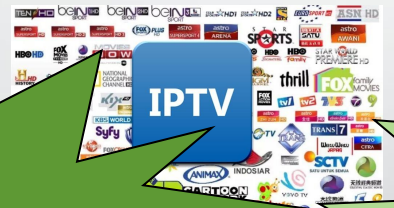


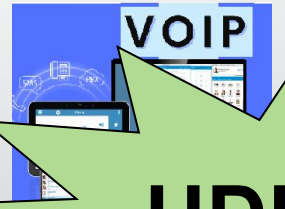
Reliable

TCP



Emails





Fast

UDP



Online Games

UDP: User Datagram Protocol [RFC 768]

- UDP : User Datagram Protocol
  - Best Effort Service
  - Used by applications that requires no delay in data delivery
- How does UDP deliver fast?
  - no connection establishment (which can add delay)
  - small header size
  - no error or flow or congestion control: UDP can blast away as fast as desired

User Datagram Protocol (UDP)

- UDP is used by:
  - streaming multimedia apps (loss tolerant, rate sensitive)
  - SNMP

  But sometimes ……
  - DNS
  - HTTP/3

- reliable transfer over UDP:
  - add reliability at application layer
  - application-specific error recovery!

Functions of the Transport Layer

- Primary responsibilities:

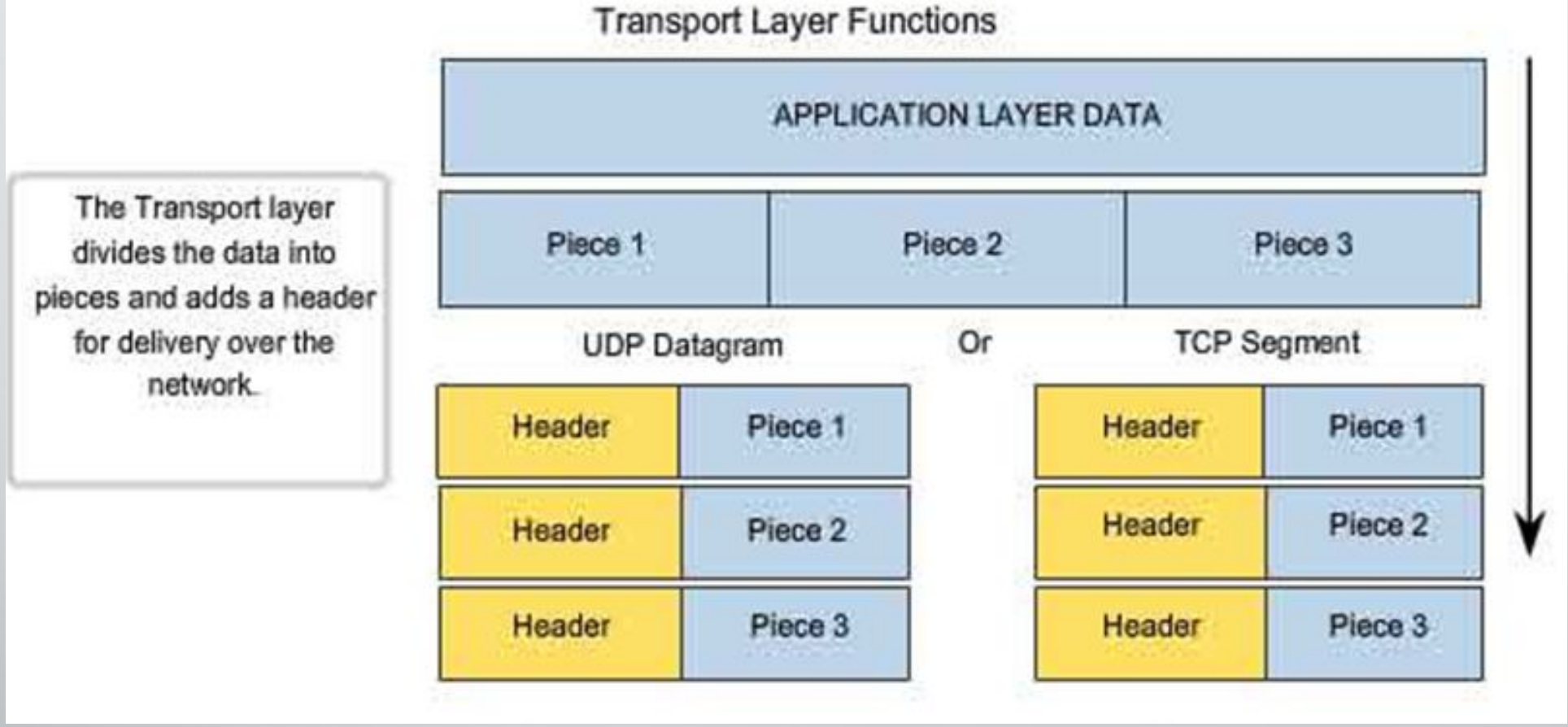
UDP &TCP

1. Segmenting the data and managing each piece.
2. Reassembling the segments into streams of application data.
3. Identifying the different applications.
4. Multiplexing

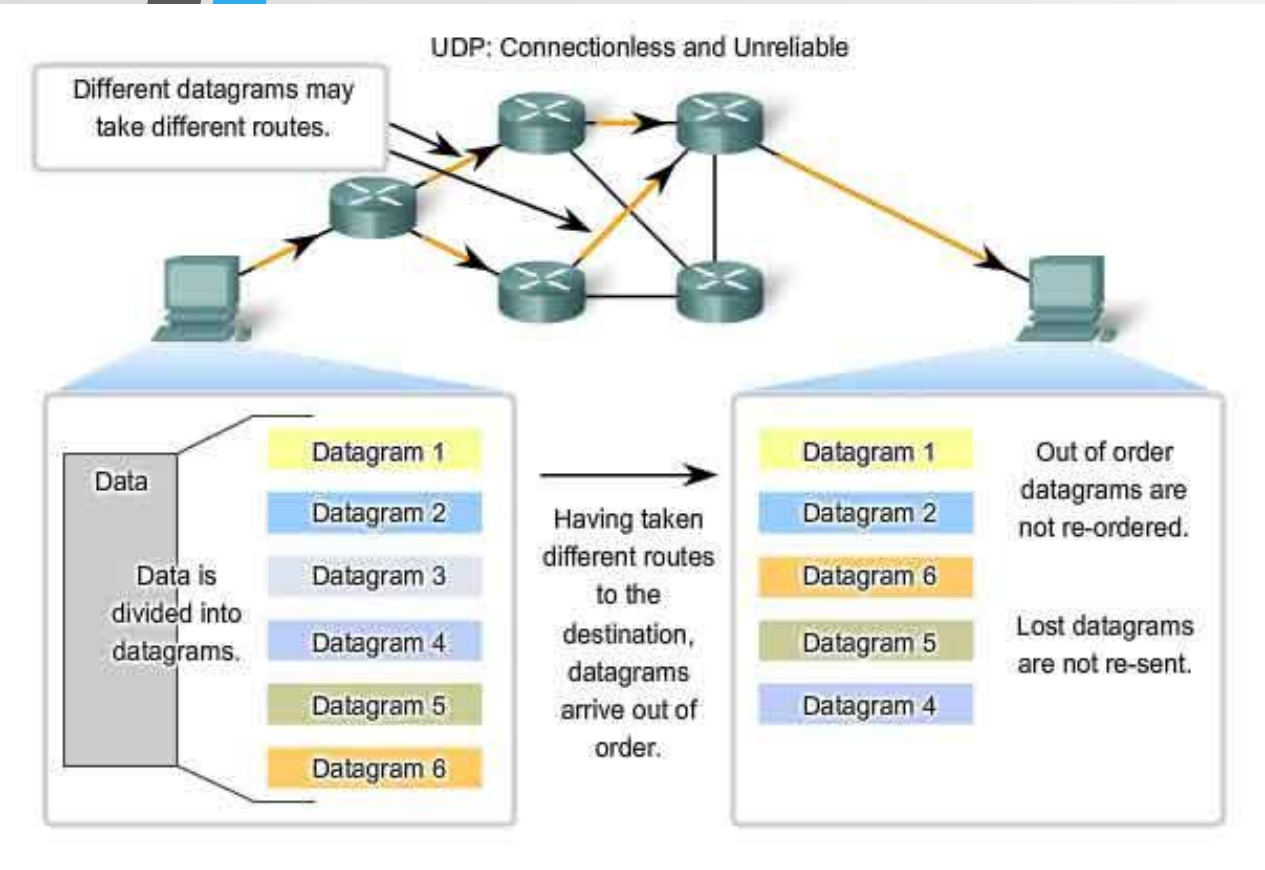
Only TCP

5. Establishing and terminating a connection
6. Enabling error recovery.
7. Performing flow control between end users.

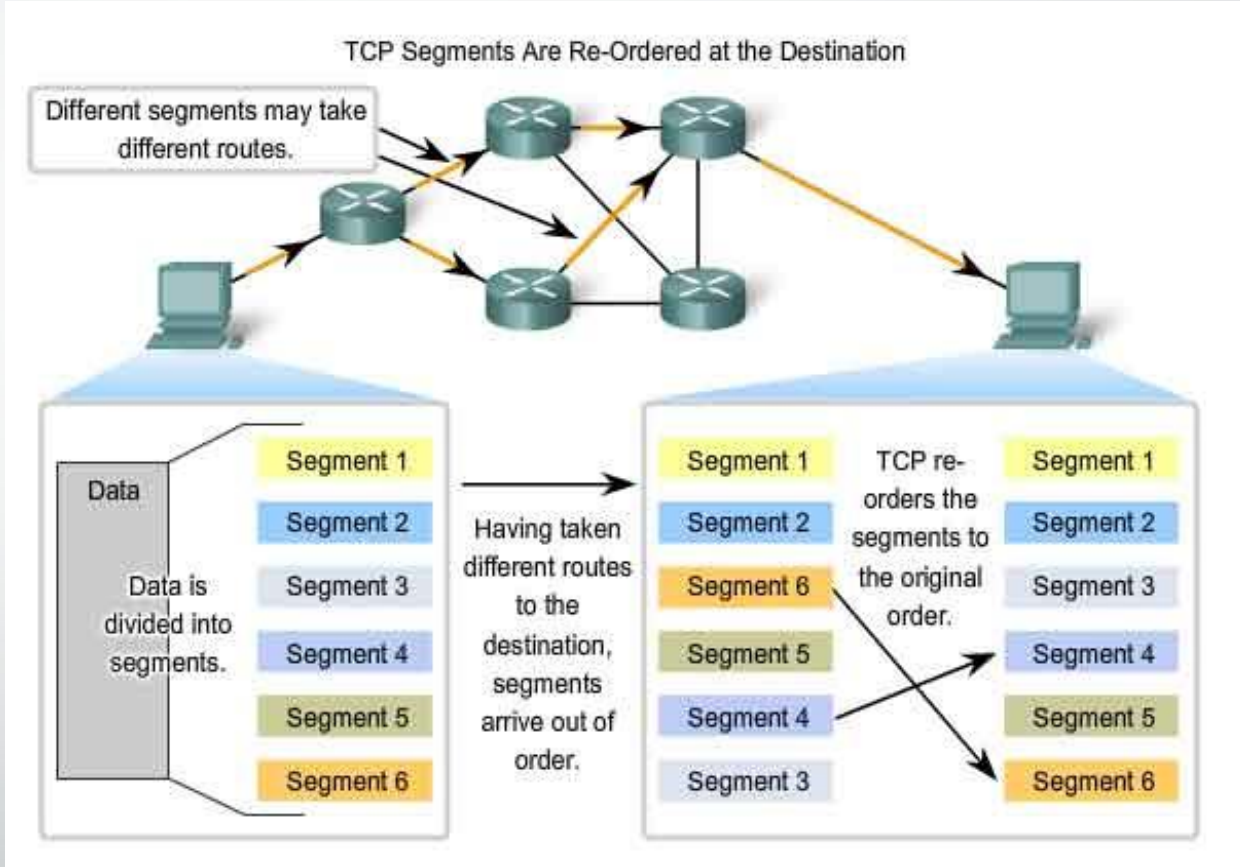
Function 1&2 – Segmentation and Reassembly



Function 2 – Reassembly



UDP



TCP

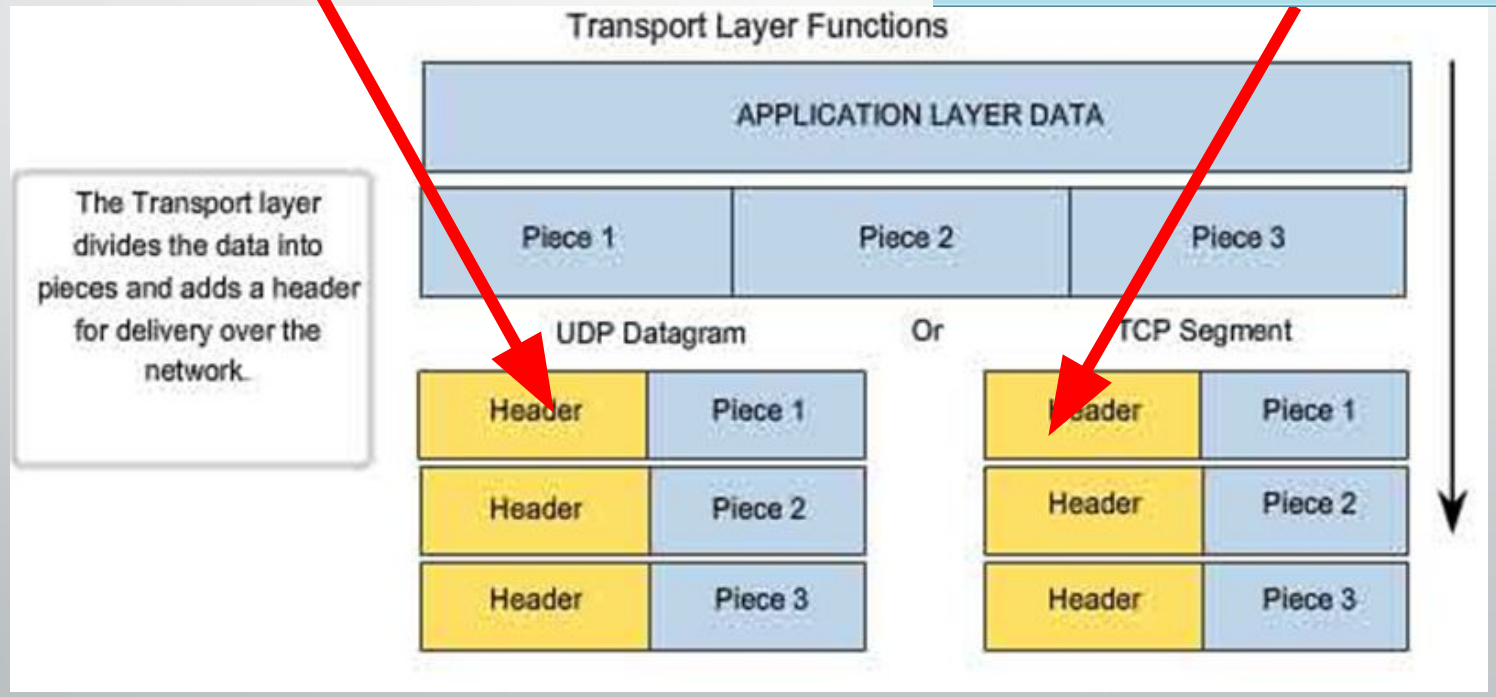
TCP and UDP Headers

UDP HEADER

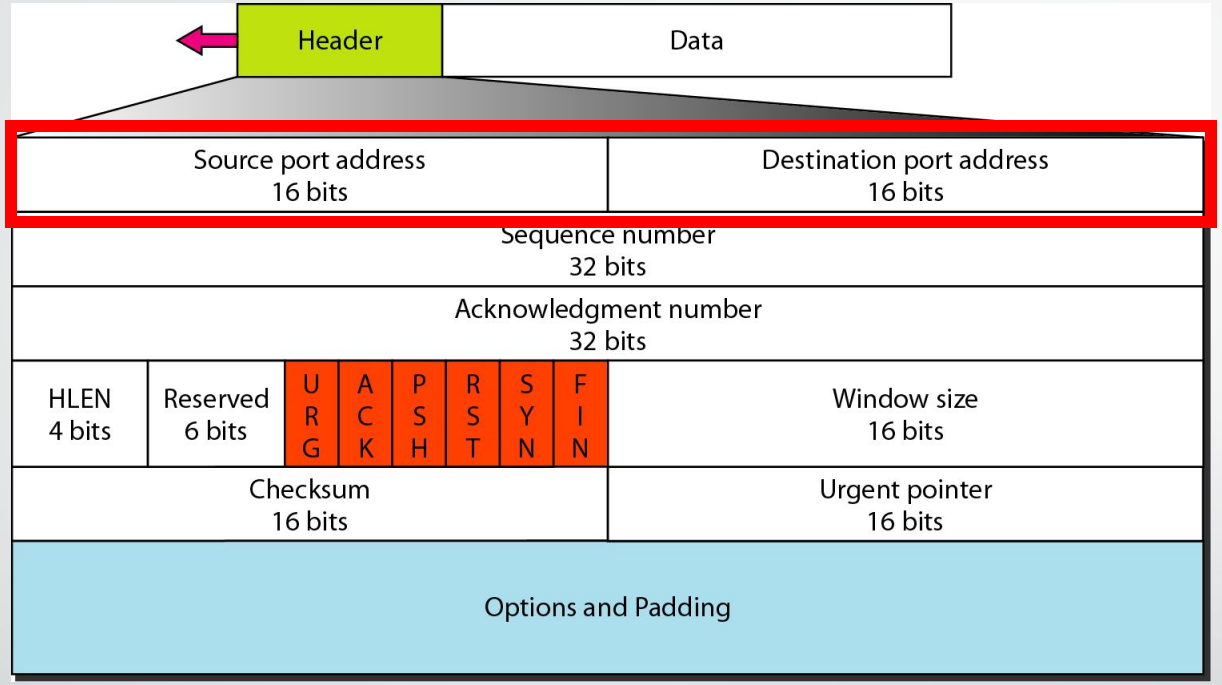
Source port number 16 bits	Destination port number 16 bits
Total length 16 bits	Checksum 16 bits

TCP HEADER

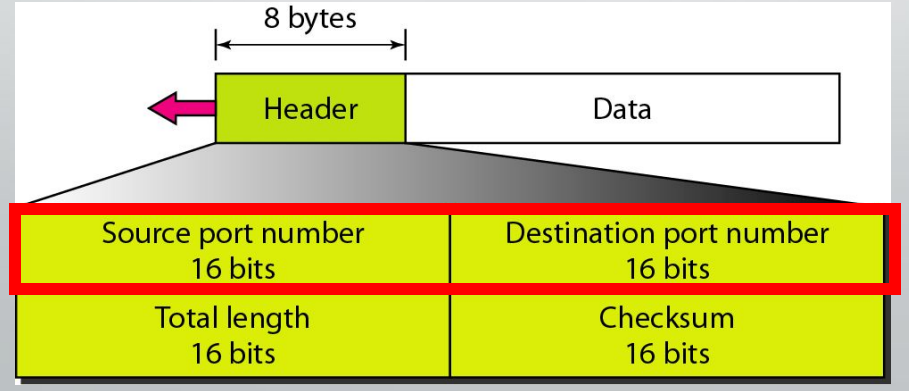
Source port address 16 bits								Destination port address 16 bits
Sequence number 32 bits								
Acknowledgment number 32 bits								
HLEN 4 bits	Reserved 6 bits	URG	ACK	PSH	RST	SYN	FIN	Window size 16 bits
Checksum 16 bits								Urgent pointer 16 bits
Options and Padding								



TCP and UDP Headers



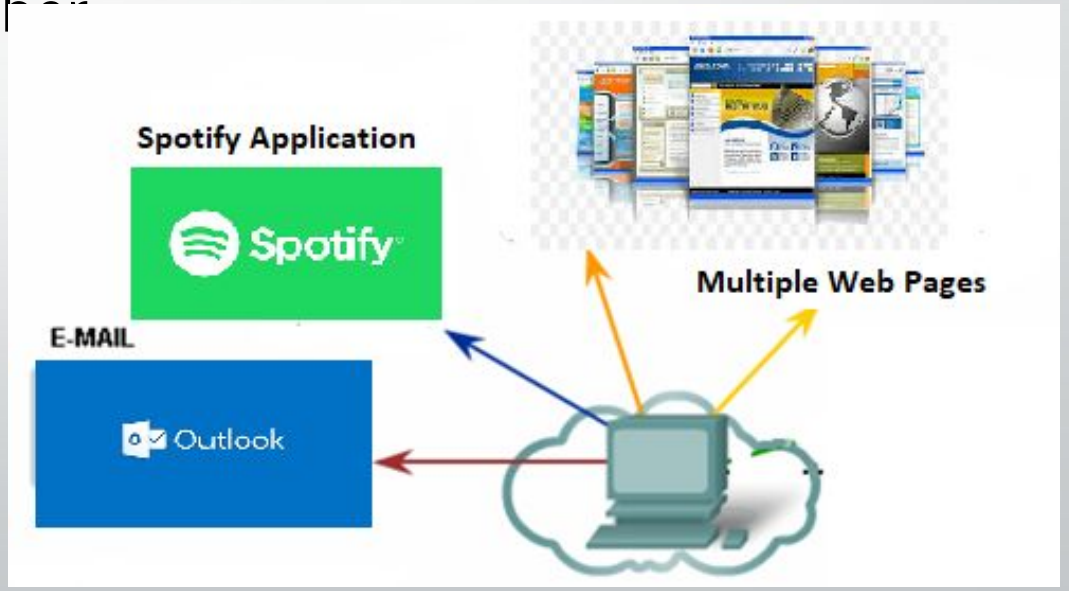
- TCP Header



- UDP Header

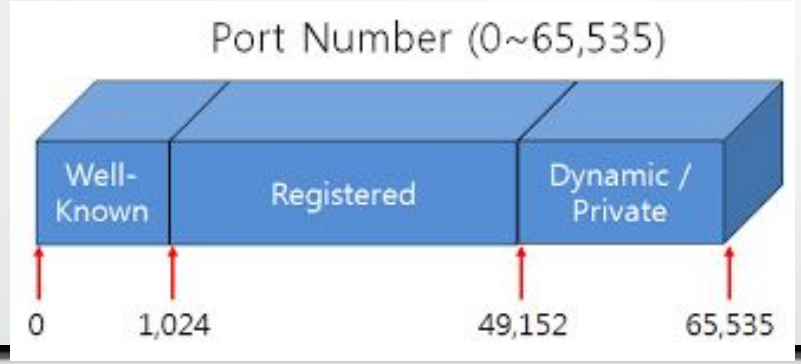
Function 3 – Identifying Different Applications

- Port Numbers/Addresses are used to identify different applications/processes running in a computer
- 16-bits in length
  - Represented as one single decimal number
  - Range **0 - 65535**
  - e.g. **80 – Web**
  - **25 – SMTP**
  - **4070 – Spotify**



Port Numbers

- Internet Corporation for Assigned Names and Numbers (ICANN) assigns port numbers
- **Three** categories:



Port Number Range	Port Group
0 to 1023	Well Known (Contact) Ports
1024 to 49151	Registered Ports
49152 to 65535	Private and/or Dynamic Ports

Port Number Types

- **Well-Known Ports:**
  - Assigned and controlled by IANA for standard services
  - Commonly used by system processes and standardized services and applications.

Port Number Range	Port Group
0 to 1023	Well Known (Contact) Ports
1024 to 49151	Registered Ports
49152 to 65535	Private and/or Dynamic Ports

67&68 – DHCP

123 – NTP

25 – SMTP

110 – POP3

143 – IMAP

443 – HTTPS

80 – HTTP

53 - DNS

Port Number Types

- **Registered Ports:**
  - Assigned by IANA but for specific applications requested by developers or organizations.
  - Can be registered for a lot of not-so-well-known, especially corporate/proprietary protocols.

Port Number Range	Port Group
0 to 1023	Well Known (Contact) Ports
1024 to 49151	Registered Ports
49152 to 65535	Private and/or Dynamic Ports

8008 – Alternate HTTP

23399 – Skype

8080 – Alternate HTTP

4070 – Spotify

5060 – SIP (VoIP)

3306 – MySQL

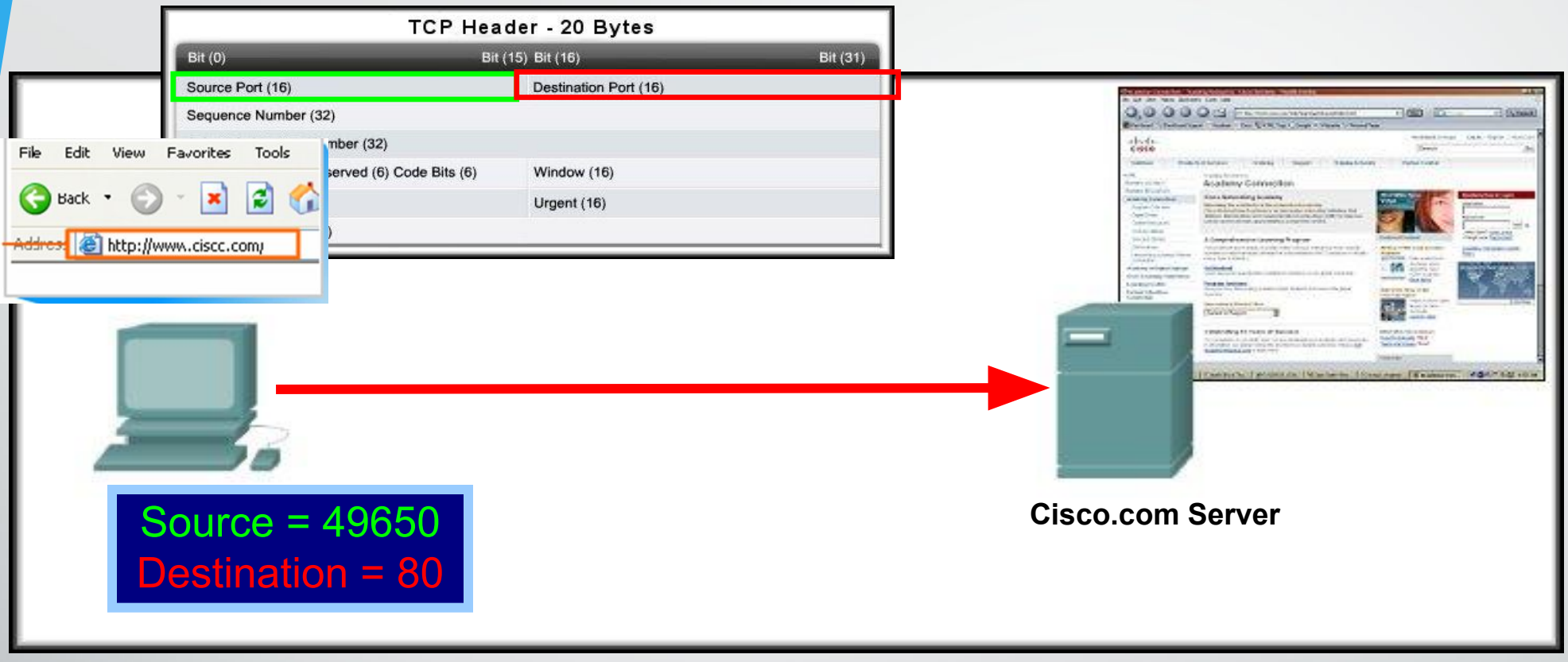
Port Number Types

- **Dynamic Ports:**
  - Also known as private or ephemeral ports
  - Never assigned or controlled by IANA.

Port Number Range	Port Group
0 to 1023	Well Known (Contact) Ports
1024 to 49151	Registered Ports
49152 to 65535	Private and/or Dynamic Ports

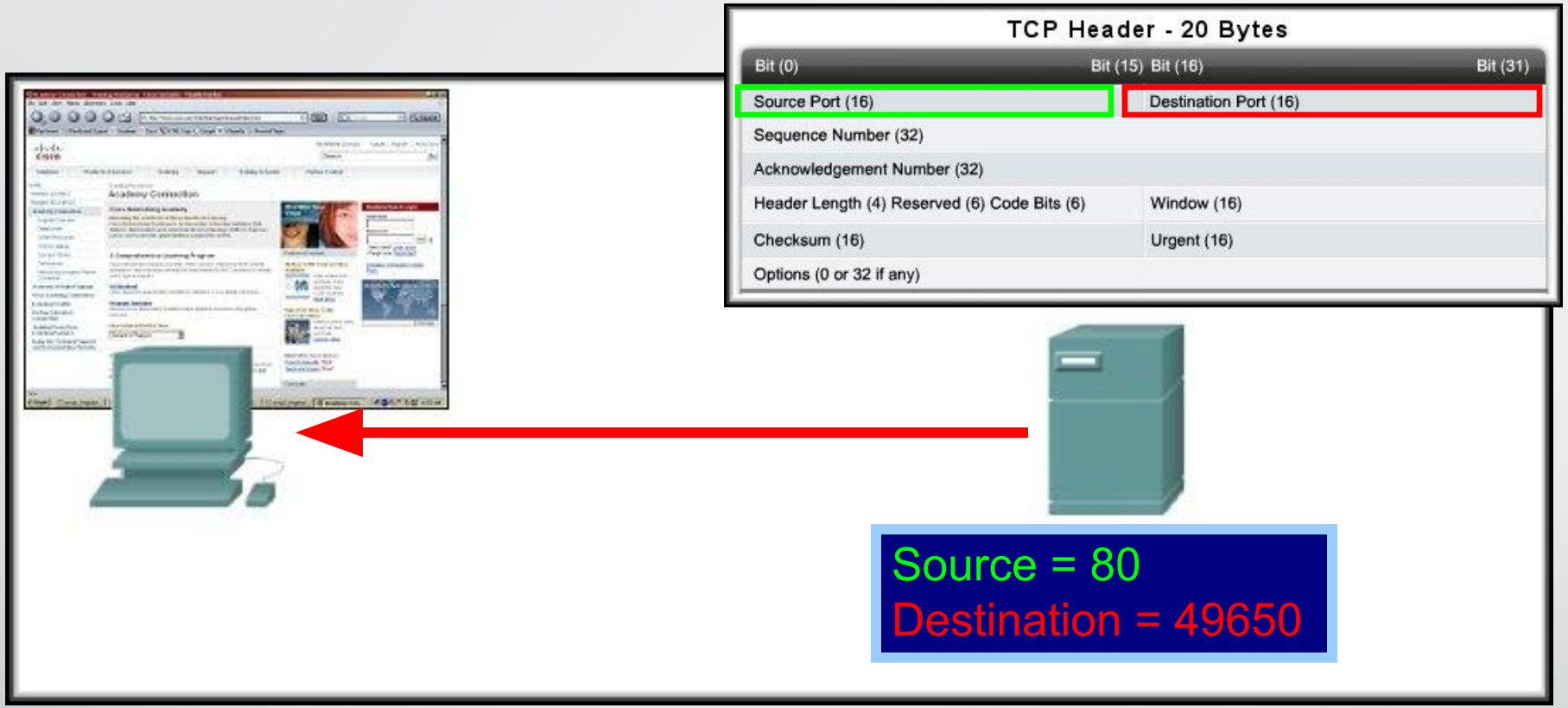
Dynamic port usage will become clearer as we move through the material.

More on Port Numbers



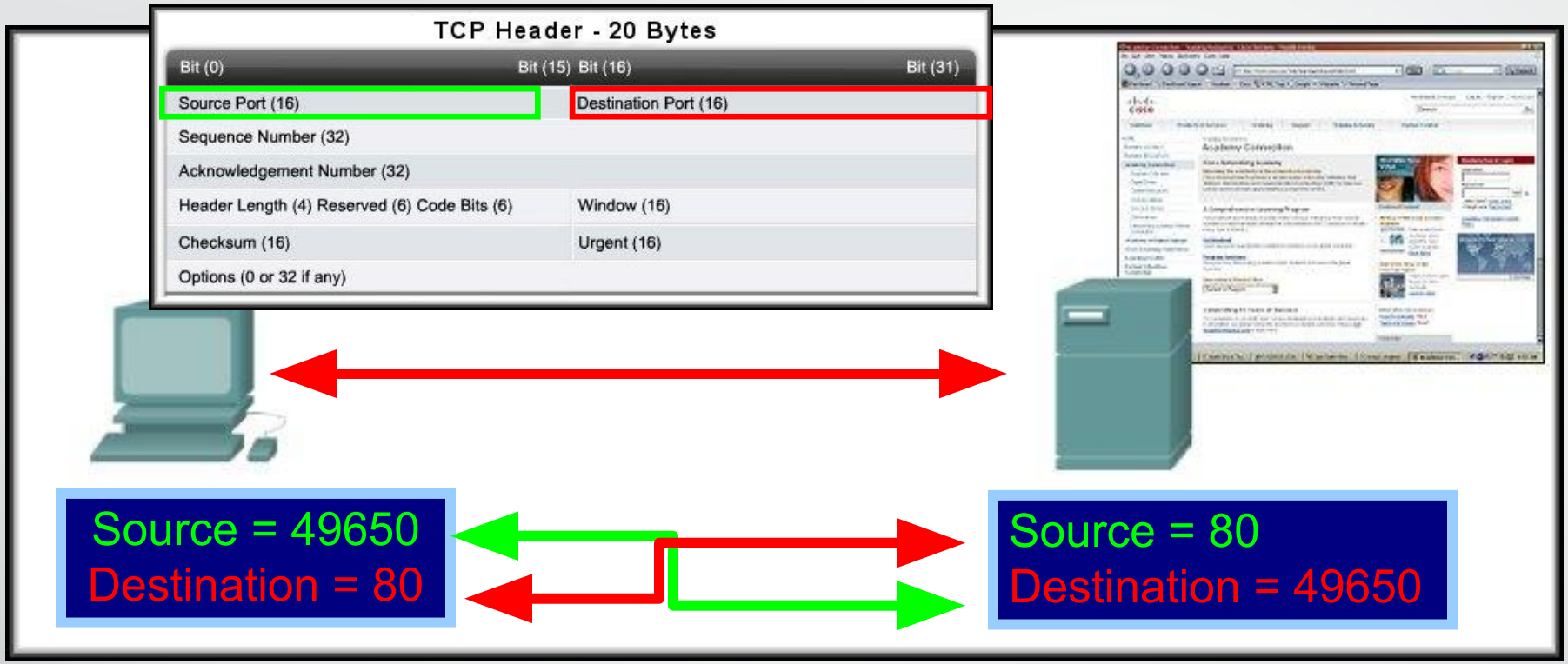
- Server is listening on Port 80 for HTTP connections.
- The client sets the destination port to 80 and uses a dynamic port as its source.

Port Numbers in Action



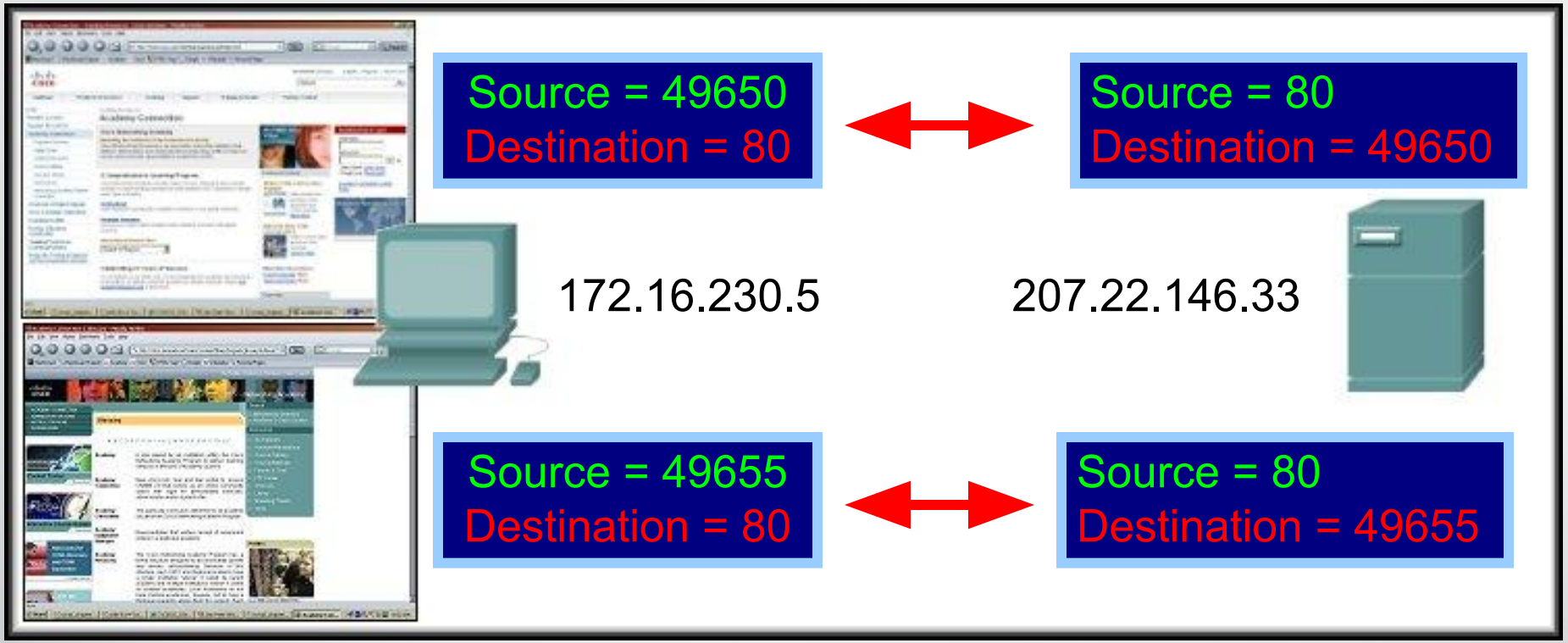
- Server replies with the web page.
  - Sets the source port to 80 and uses the client's source port as the destination.

Port Numbers in Action



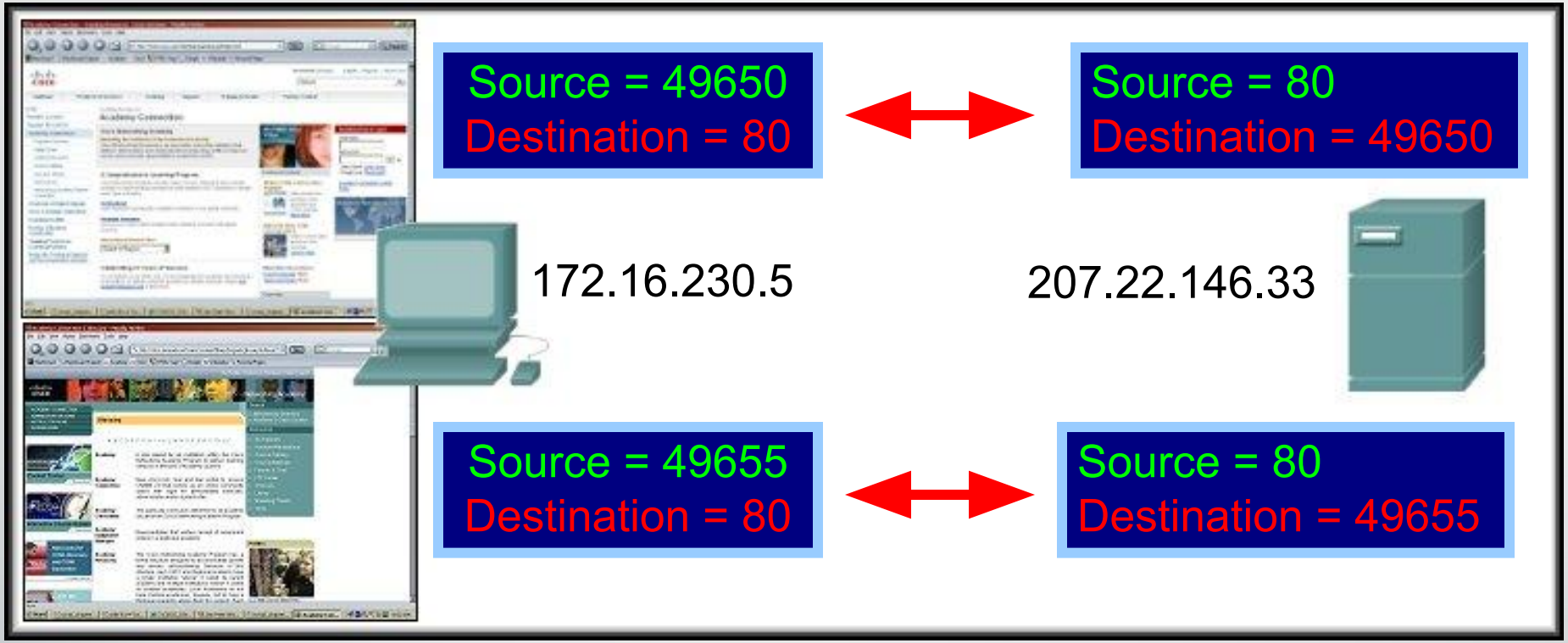
- Clients can use any random port number, Servers can't.
  - Because clients won't be able to identify server process otherwise
  - Servers thus must use **well-known port numbers**!

Port Numbers in Action



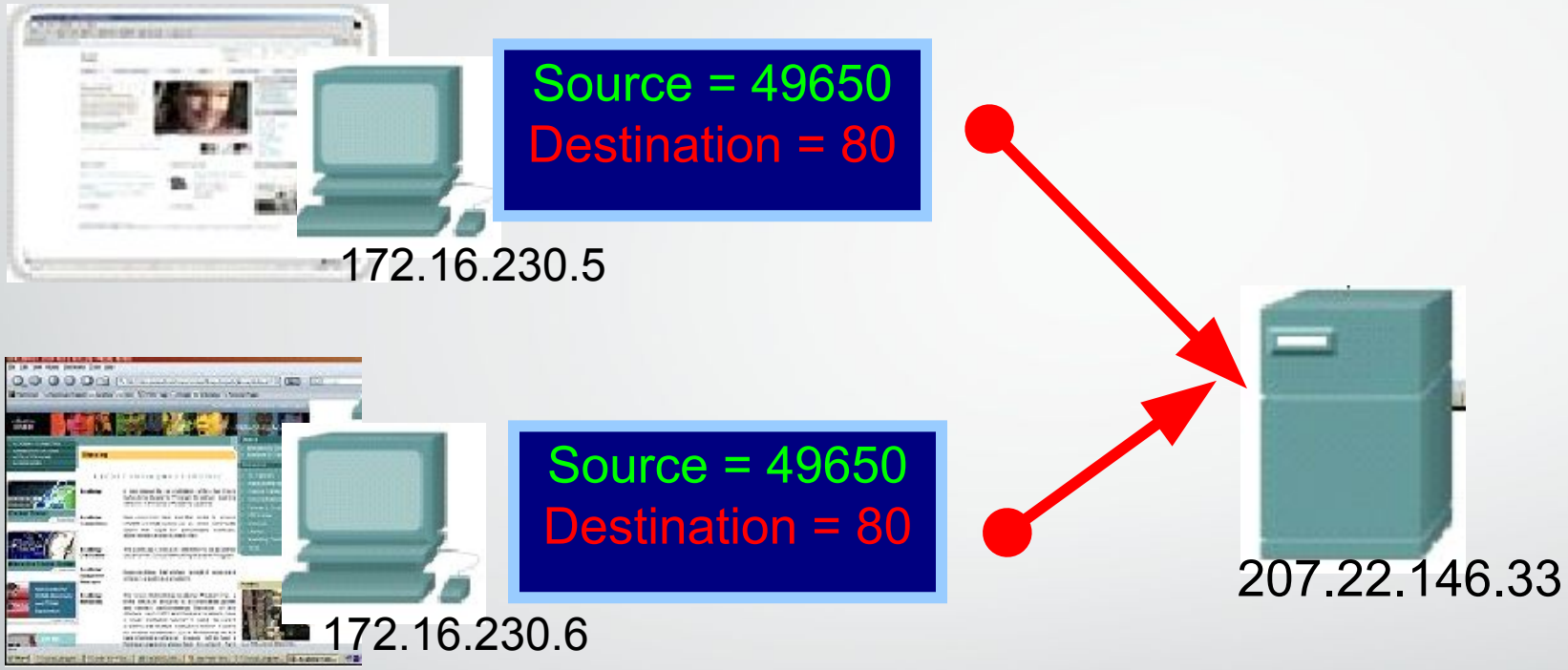
- What if there are two sessions to the same server?
  - The client uses another dynamic port as its source and the destination is still port 80.
  - Different source ports keep the sessions unique on the server.

Port Numbers in Action



- There are two tabs in the same PC, then?
  - The client uses another dynamic port as its source and the destination is still port 80.
  - Different source ports keep the sessions unique.

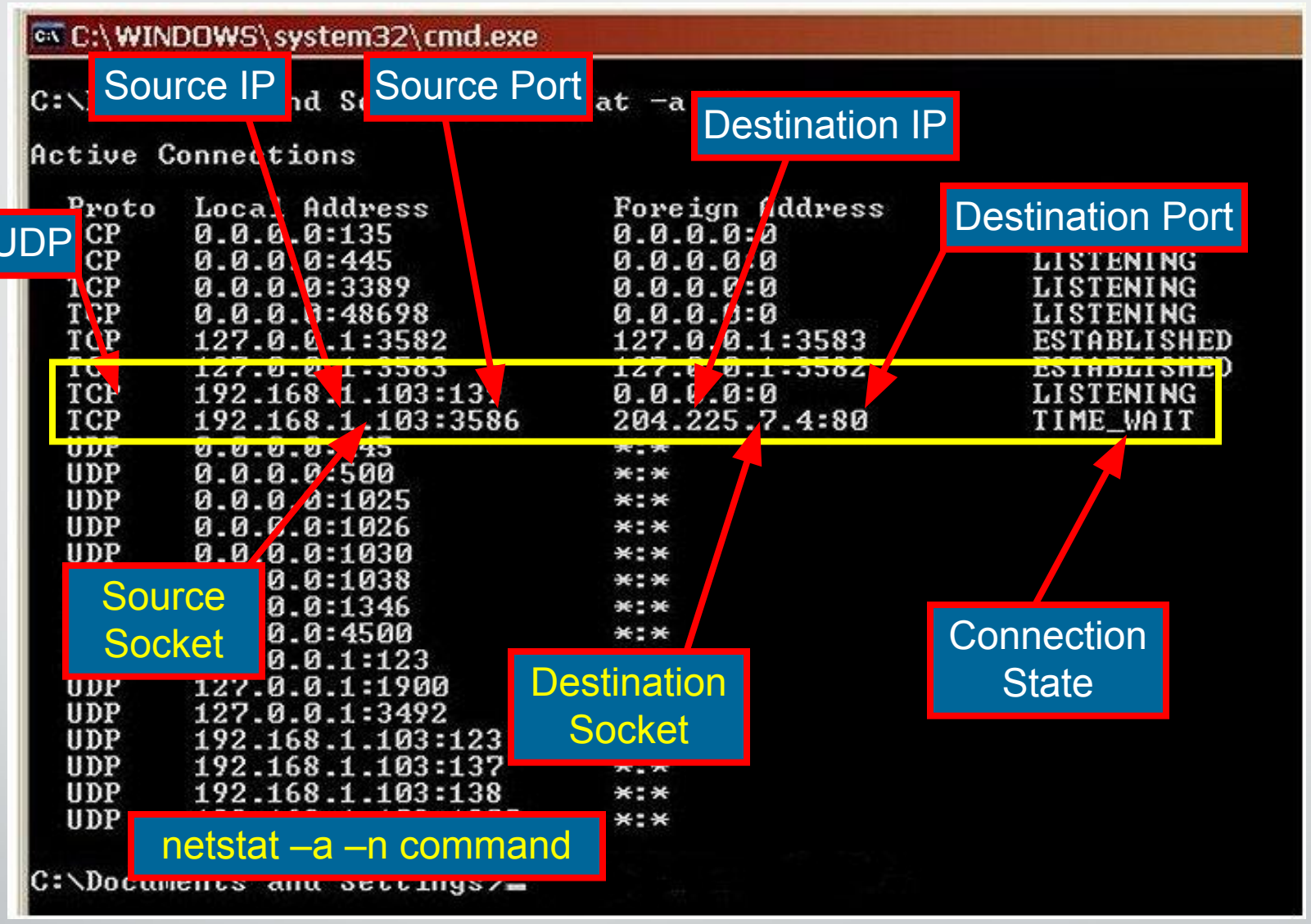
More on Port Numbers in Action



- How does the Server's Transport Layer keep them separate?
  - The socket (IP Address:Port)

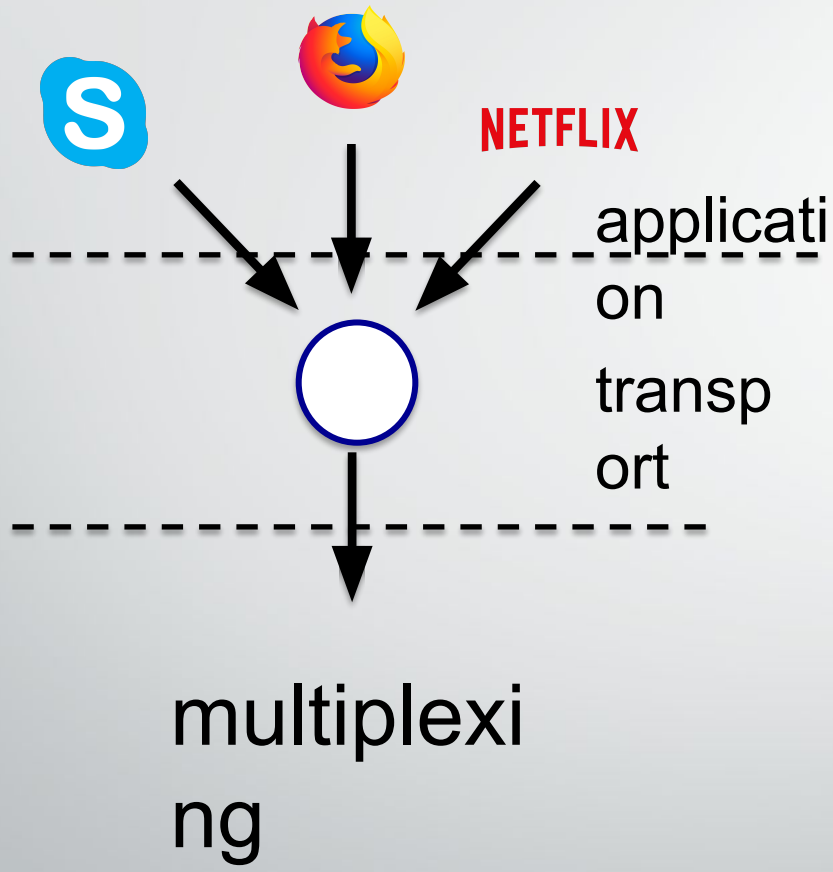
172.16.230.5:49650 ⟷ **207.22.146.33:80**

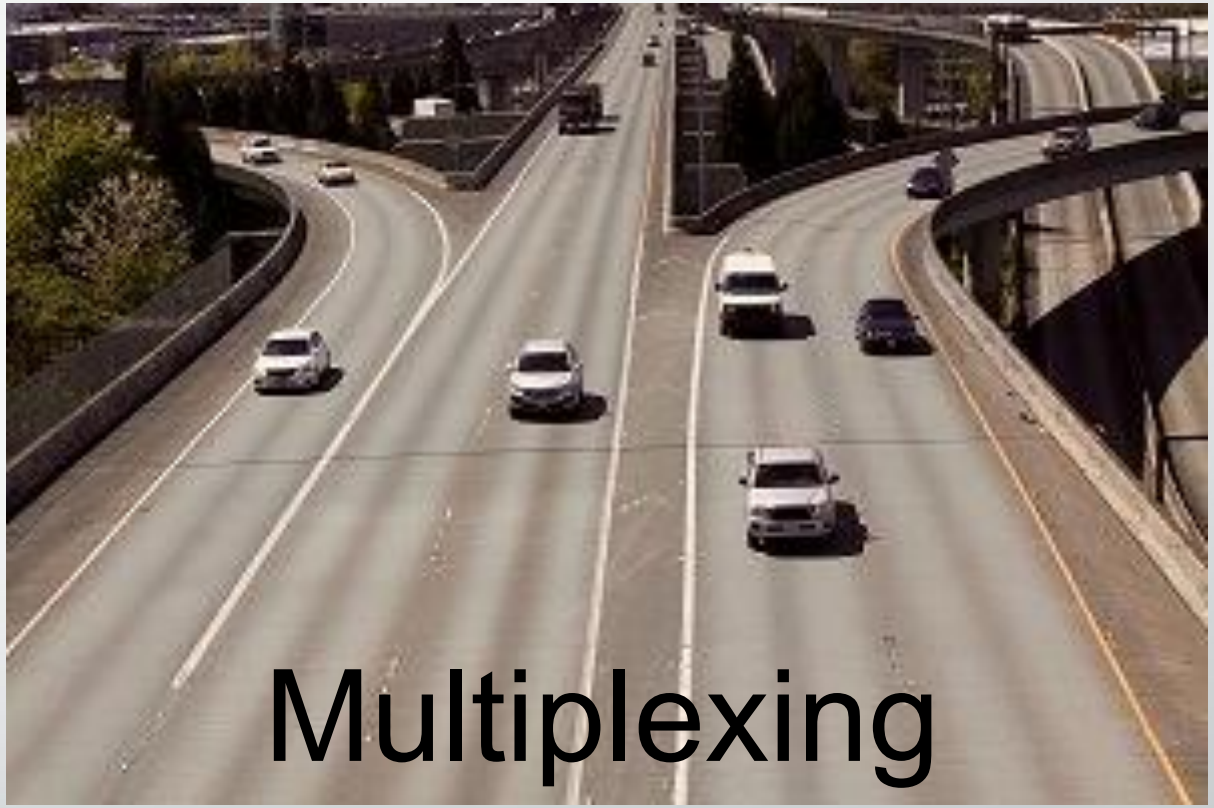
172.16.230.6:49650 ⟷ **207.22.146.33:80**



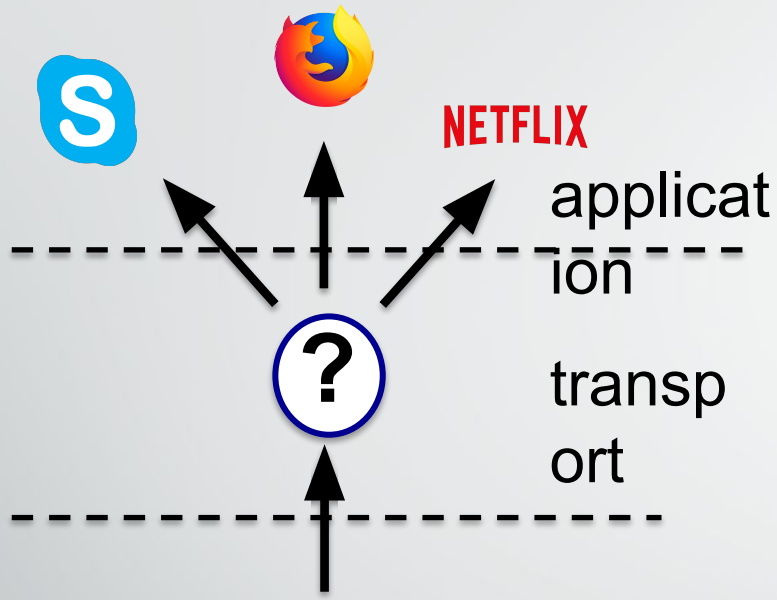
Netstat - Network Utility Tool

Function 4 –Multiplexing





Function 4 –DeMultiplexing



Demultiplexing



Demultiplexing

DeMultiplexing/ Multiplexing

- Multiplexing, demultiplexing: based on segment, datagram header field values
- **UDP:** demultiplexing using **destination port number** (only)
- **TCP:** demultiplexing using 4-tuple: **source and destination IP addresses, and port numbers**

And Now more on TCP!